

Austin Hutchen

I'm a Computer Science student at the University of Colorado Boulder who specializes in building cross-platform software, e-commerce websites, SEO, digital marketing, and crafting embedded scientific instruments.

 Boulder, CO
  hutchencodes@gmail.com
  austinmorozane.github.io/austinscode
  austin-h-15440a1b2
 austinmorozane

Education

- | | |
|---|------------------------------------|
| BS University Of Colorado Boulder , Computer Science | Boulder, CO
Aug 2023 – May 2028 |
| <ul style="list-style-type: none"> GPA: 3.097 Built several projects including a quartz-powered oscillator clock, LED animation wall, automatic plant watering system, user vibe-based location search API, calculator app, note app, and more. Check out github and my website for the full list of (35) projects. | |
| BS University Of Colorado Boulder , Mathematics | Boulder, CO
Aug 2026 – May 2028 |
| <ul style="list-style-type: none"> GPA: 3.17 Recieved letter of recommendation for honors working as Calculus 2 Course Assistant. | |
| MN University Of Colorado Boulder , Computer Engineering | Aug 2026 – May 2028 |

Experience

- | | |
|--|--|
| FormFactor , Software Engineering Intern
Working as an embedded software engineer , collaborated with engineering team and wrote tests for a firmware simulator built on the MODBUS TCP standard. | Gunbarrel, CO
May 2024 – Aug 2024
4 months |
| <ul style="list-style-type: none"> Developed a comprehensive ModBus ASCII/RTU library in C#, including simulators and extensive tests. Designed and tested a NUGET driver for interfacing with an RS-485 TCP server and an embedded temperature controller instrument (MOXA CND3) used in a quantum computer for near-zero Kelvin operations. Collaborated directly with the FormFactor Cryo Boulder team on the IQ-3000 Spin-Qubit Quantum Computer system. Using documentation, optimized ASCII/RTU algorithm for efficient retrieval of data on slower networks over TCP. | |
| University Of Colorado Boulder , Course Assistant
Taught alongside faculty and held office hours weekly throughout the semester, to assist students struggling in APPLIED and MATH 2400 at CU Boulder. | Boulder, CO
Aug 2023 – Dec 2023
5 months |
| <ul style="list-style-type: none"> Developed visualization strategies for solving complex integrals, such as u-sub, volumes of solids, series, and inverse functions. Assisted with in-class recitations, in which students solved worksheets of related chapter problems weekly. Went to weekly meetings with the MATH department to discuss approaches for helping students with common problems in the course. | |

Mobius Materials, Digital Marketing Analyst

Worked as leader of marketing team, viewing weekly analytics and using them to further the development of online marketplace and quality of marketing material.

San Francisco, CA
May 2021 – Aug 2021
4 months

- Developed an article about the importance of regulating and developing tools to reuse e-waste [Why Sustainability Ultimately Begins With E-Waste, Not Paper Straws](#), which led to the company's first major sale of repurposed components.
- Assisted Software Engineering team with development of e-waste marketplace UI.
- Led e-mail marketing campaigns, and wrote several drafts articulating the importance of a circular economy to interested buyers.

Projects

TheVibeCheck @

Sept 2024 – Nov 2024

Open-source application for returning nearby locations based on the user's inputted mood. Working on a team, I integrated the google maps Places API, managed the CSS, and developed the main scripts.

- Achieved 2.8x speedup over baseline application launch on slower wifi connections by helping to optimize the SQL backend and CSS.
- Adopted by 3 contributors

WebCam @

June 2024 – Sept 2024

A webcam application similar to photobooth that captures realtime video, and compresses it using the BLOB format. Allows for upload to an Amazon FireBase database for storage and retrieval post-compression.

- Reduced streaming lag and file size by 50% with less than 10% accuracy degradation from initial release

Calculator App @

July 2021 – June 2023

An IOS calculator app built from scratch using C and Flutter, that supports transcendental functions. The UI was designed to look familiar, similar to the default IOS calculator app.

- Added optimization helpers for functions like $\sqrt{x}$, $\sin(x)$, $\arctan(x)$, π , and similar operations.
- Reduced bundle size by 30% by using a main switch function, and optimized helpers for more complex operations.

Skills

Languages: C#, TypeScript, JavaScript, CSS, Java, SQL, Python, C++, CUDA, Rust, Julia

Infrastructure: Kubernetes, Docker, FireBase, AWS

Research Areas: Embedded Software Engineering, Firmware Development, Software Development, Computer Systems, Mathematics, and Mobile Development

Invited Talks

1. Group Theory, Abstract Algebra, and teaching complex Mathematics [The Quaternionic Learning Process](#)